CHUIYANG MENG

IEEE Member ⋄ chuiyang_meng@sfu.ca

Homepage LinkedIn Google Scholar

RESEARCH INTERESTS

Foundation Models, Mechanistic Interpretability, Multi-Agent Systems, Distributed Learning, Communication Networks.

EDUCATION

The University of British Columbia, Vancouver, Canada *September 2021 - June 2026*
Doctor of Philosophy (Ph.D.), Department of Electrical and Computer Engineering
Supervisor: Vincent W.S. Wong
Thesis: Resource-aware Distributed Edge Intelligence

Johns Hopkins University, Homewood, Baltimore, U.S.A. *August 2018 - May 2020*
Master of Science (M.S.), Department of Electrical and Computer Engineering

Harbin Institute of Technology, Harbin, P.R.C. *September 2014 - June 2018*
Bachelor of Engineering (B.Eng.), School of Electronics and Information Engineering
Supervisors: Shuai Han and Hsiao-Hwa Chen

EXPERIENCE

Simon Fraser University, Burnaby, Canada *September 2026 - Present*
Postdoctoral Research Fellow, School of Engineering Science
Supervisor: Cheng Li

Vector Institute, Toronto, Canada *January 2026 - April 2026*
Applied Machine Learning Intern, Responsible AI Team

- Developed deepfake video detection methods for identifying manipulated content.
- Built training and validation pipelines for video forgery detection.

The University of British Columbia, Vancouver, Canada *September 2021 - June 2026*
Graduate Research Assistant

- Developed federated LLM fine-tuning with low-rank Gram-matrix aggregation and Procrustes alignment (FLoRG).
- Developed zeroth-order federated LLM fine-tuning with heterogeneous block activation (ZorBA).
- Designed adaptive model splitting and online resource allocation for split federated learning (ASFL).
- Developed graph neural network methods for neighbor selection and resource allocation in decentralized federated learning.

PUBLISHED AND ACCEPTED PUBLICATIONS

Conference Publications

1. **Chuiyang Meng**, Ming Tang, and Vincent W.S. Wong. "FLoRG: Federated Fine-tuning with Low-rank Gram Matrices and Procrustes Alignment." In *Proceedings of the International Conference on Learning Representations (ICLR)*, Rio de Janeiro, Brazil, Apr. 2026.

2. **Chuiyang Meng**, Ming Tang, and Vincent W.S. Wong. "ZorBA: Zeroth-order Federated Fine-tuning of LLMs with Heterogeneous Block Activation." In *Proceedings of the IEEE International Conference on Computer Communications (INFOCOM)*, Tokyo, Japan, May 2026.

3. **Chuiyang Meng**, Ming Tang, Mehdi Setayesh, and Vincent W.S. Wong. "GNN-based Neighbor Selection and Resource Allocation for Decentralized Federated Learning." In *Proceedings of the IEEE Global Communications Conference (GLOBECOM)*, Kuala Lumpur, Malaysia, pp. 1223–1228, Dec. 2023.

4. Xiao Deng, Xuanli Wu, **Chuiyang Meng**, Shuai Han, and Xiaojie Fang. "A Mobile Relay Selection Strategy in Cooperative Spectrum Sharing Framework." In *Proceedings of the IEEE International Conference on Communications (ICC)*, Paris, France, pp. 1–6, May 2017.

Journal Publications

1. **Chuiyang Meng**, Ming Tang, and Vincent W.S. Wong. "ASFL: An Adaptive Model Splitting and Resource Allocation Framework for Split Federated Learning." *IEEE Transactions on Mobile Computing (TMC)*, vol. 25, no. 10, pp. 18484–18501, Oct. 2026.

2. **Chuiyang Meng**, Ming Tang, Mehdi Setayesh, and Vincent W.S. Wong. "Tackling Resource Allocation for Decentralized Federated Learning: A GNN-based Approach." *IEEE Transactions on Mobile Computing (TMC)*, vol. 24, no. 10, pp. 9554–9569, Oct. 2025.

3. Zijun Gong, Cheng Li, Fan Jiang, Ruoyu Su, Ramachandran Venkatesan, **Chuiyang Meng**, Shuai Han, Yan Zhang, Shudong Liu, and Kun Hao. "Design, Analysis, and Field Testing of an Innovative Drone-Assisted Zero-Configuration Localization Framework for Wireless Sensor Networks." *IEEE Transactions on Vehicular Technology (TVT)*, vol. 66, no. 11, pp. 10322–10335, Nov. 2017.

4. Cheng Li, Fan Jiang, **Chuiyang Meng**, and Zijun Gong. "A New Turbo Equalizer Conditioned on Estimated Channel for MIMO MMSE Receiver." *IEEE Communications Letters*, vol. 21, no. 4, pp. 957–960, Apr. 2017.

MANUSCRIPTS UNDER REVIEW

1. **Chuiyang Meng**, Wenlu Yu, Ming Tang, and Cheng Li. "Social Circuits Behind Multi-Agent Echo Chambers." Manuscript, 2026. Submitted to ICLR 2027.

2. **Chuiyang Meng**, Ming Tang, and Vincent W.S. Wong. "CORTEX: Learning to Share and Specialize in Dense Language Models." Manuscript, 2026. Submitted to ICLR 2027.

TALKS

1. "Rethinking Collaboration: Cost and Value in Fine-Tuning and Multi-Agent Reasoning." IEEE MWN Workshop on 6G for Ubiquitous Intelligence, University of Victoria, Victoria, Canada, Nov. 11, 2026 (*upcoming*).

PROFESSIONAL SERVICES

Symposium Co-Chair

– International Conference on Wireless Communications and Signal Processing (WCSP), Shenzhen, China, 2026.
Symposium 6: Edge Intelligence, Computing, and Caching.

Conference Reviewer

– Conference on Neural Information Processing Systems (NeurIPS)

– IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)

– International Conference on Learning Representations (ICLR)

Journal Reviewer

– IEEE Transactions on Wireless Communications (TWC)

– IEEE Transactions on Mobile Computing (TMC)

– IEEE Transactions on Communications (TCOM)

– IEEE Internet of Things Journal (IoT-J)

– IEEE Wireless Communications Magazine

AWARDS

– IEEE INFOCOM 2026 Student Travel Grant ($1640 USD)

– President's Academic Excellence Initiative PhD Award, The University of British Columbia, 2021 - 2026

– International Tuition Award, The University of British Columbia, 2021 - 2026

TEACHING

Teaching Assistant

– **ELEC 331 Computer Communications**
Fall 2021, Spring 2022, Fall 2023, Spring 2024, Spring 2025
The University of British Columbia, Department of Electrical and Computer Engineering

– **EN 520.647 Information Theory**
Fall 2019
Johns Hopkins University, Department of Electrical and Computer Engineering